Experiment No:17
Experiment to understand Linear Regression For a given data set.

AIM:
To understand Linear Regression for a given data set.

DESCRIPTION:
Understand the Linear regression for the dataset given.

CODE:

```
import numpy as np
import pandas as pd

df=pd.read_csv('/content/student_data.csv')
df
df.info()
df.dropna(inplace=True)
df.info()
df.describe()

features=df.iloc[:,[2]].values
label=df.iloc[:,[32]].values

from sklearn.model_selection import train_test_split
x_train,x_test,y_train,y_test=train_test_split(features,label,te
st_size=0.2,random_state=42)

from sklearn.linear_model import LinearRegression
model=LinearRegression()
model.fit(x_train,y_train)
model.score(x_train,y_train)

model.score(x_test,y_test)
```

```
model.coef_

model.intercept_

import pickle
pickle.dump(model,open('SalaryPred.model','wb'))
model=pickle.load(open('SalaryPred.model','rb'))

age=float(input("Enter Student Age: "))
age_NP=np.array([[age]])
grade=model.predict(age_NP)

print("Predicted final grade for age {} is
{}".format(age,grade))
```

OUTPUT:

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 395 entries, 0 to 394
Data columns (total 33 columns):
 #   Column      Non-Null Count  Dtype
---  ------      --------------  -----
 0   school      395 non-null    object
 1   sex         395 non-null    object
 2   age         395 non-null    int64
 3   address     395 non-null    object
 4   famsize     395 non-null    object
 5   Pstatus     395 non-null    object
 6   Medu        395 non-null    int64
 7   Fedu        395 non-null    int64
 8   Mjob        395 non-null    object
 9   Fjob        395 non-null    object
 10  reason      395 non-null    object
 11  guardian    395 non-null    object
 12  traveltime  395 non-null    int64
 13  studytime   395 non-null    int64
```

```
 14  failures    395 non-null    int64
 15  schoolsup   395 non-null    object
 16  famsup      395 non-null    object
 17  paid        395 non-null    object
 18  activities  395 non-null    object
 19  nursery     395 non-null    object
 20  higher      395 non-null    object
 21  internet    395 non-null    object
 22  romantic    395 non-null    object
 23  famrel      395 non-null    int64
 24  freetime    395 non-null    int64
 25  goout       395 non-null    int64
 26  Dalc        395 non-null    int64
 27  Walc        395 non-null    int64
 28  health      395 non-null    int64
 29  absences    395 non-null    int64
 30  G1          395 non-null    int64
 31  G2          395 non-null    int64
 32  G3          395 non-null    int64
dtypes: int64(16), object(17)
memory usage: 102.0+ KB
<class 'pandas.core.frame.DataFrame'>
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 29  absences    395 non-null    int64
 30  G1          395 non-null    int64
 31  G2          395 non-null    int64
 32  G3          395 non-null    int64
dtypes: int64(16), object(17)
memory usage: 102.0+ KB
Enter Student Age: 44
Predicted final grade for age 44.0 is [[-4.07873452]]
```

RESULT:

Thus, Linear Regression was successfully implemented on the student dataset to predict the final grade (`G3`) based on the student's age. The model was trained and tested using the `train_test_split()` method, and its performance was evaluated using the score function. The trained model was saved and loaded using Pickle, and the predicted final grade was obtained for the given age.